UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS



Doctoral Program in Deep Tech, Artificial Intelligence & Emerging Technologies

Research Methods and Scientific Integrity in AI and Advanced Technologies

DELIVERABLE No. 6: Reproducibility Audit of a Published Paper

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Reproducibility Audit Report

Reproducibility audit of a published paper

Audited paper: Alenezi, M., Akour, M., & Al-Omari, O. (2026). Agentic artificial intelligence in software engineering: Integration points, architectural frameworks, empirical evaluation, and governance. *International Journal of Technology, 17*(4), 1245–1262. https://doi.org/10.14716/ijtech.v17i4.8380

Overall score: 3.5 / 9 (≈ 3.9 / 10), low reproducibility. The checklist applied is the course *Reproducibility Scorecard* (Yes / Partial / No); the machine-learning items were adapted to the quasi-experimental design of the paper.

1. Paper Metadata

Alenezi et al. (2026) combine a systematic literature review of 34 studies with a quasi-experimental before–after analysis of four enterprise teams; the study is auditable, although its sample is limited to N = 4.

Field	Detail
Citation	Alenezi, M., Akour, M., & Al-Omari, O. (2026). *International Journal of Technology, 17*(4), 1245–1262.
DOI	https://doi.org/10.14716/ijtech.v17i4.8380
Type and dates	Research article; received January 2026, revised February 2026, accepted May 2026.
Affiliations	Saudi Data & AI Authority; Al Yamamah University; Prince Sultan University (Riyadh).
Research questions	(1) Which SDLC phases yield the greatest benefit; (2) which architectural and governance patterns enable safe deployment; (3) measurable impact on quality, productivity, and reliability; (4) comparison with a baseline without AI.
Design	Systematic literature review (2024–2025, 34 studies) plus a multi-case quasi-experimental pre/post study, April–September 2025.
Main result	Reductions of 40–55 % in lead/cycle time and 33–48 % in MTTR, higher test coverage, fewer change failures, and no compliance breaches.
Stated limitations	Small sample, single organization, no randomization, 12-week observation window.
Funding, conflicts, AI use	Article processing charge (APC) paid by Prince Sultan University; no conflicts of interest; the authors state that no AI tools were used for the text, results, or statistical analysis.
Code and data	No availability statement.

2. Experimental Setup

The design is a within-team before–after comparison: an 8-week baseline and a 12-week treatment period across four teams, with no control group and no randomization (Sec. 3.4).

Project	Team	Codebase and stack	Agent function (risk)
A. Banking (API)	11	2.1 M LOC; Java, Spring, Kubernetes	Code generation, refactoring, testing (high, PII)
B. Health (claims)	9	~850 K LOC; Python, FastAPI, FHIR	Test synthesis, compliance (high, PHI)
C. Analytics	12	1.3 M LOC; Scala, Spark, Airflow	ETL migration, documentation (medium)
D. Platform engineering	10	~650 K LOC; Go, Terraform, Prometheus	Incident triage, remediation (medium)

- **Intervention:** a shared multi-agent platform (planning, retrieval-augmented code generation, testing, review, CI/CD, compliance, observability) integrated with Jira, GitLab, JUnit/pytest, SonarQube/ZAP, Terraform, Prometheus/Grafana, and Open Policy Agent. Neither the product nor the model is named.

- **Variables:** twelve variables across five dimensions (Table 2)—process, quality, reliability, security/compliance, and cost—measured weekly from Git, CI/CD, issue trackers, SonarQube, and observability data, plus optional weekly developer surveys.
- **Analysis:** Wilcoxon signed-rank test ($\alpha = 0.05$), Cliff's δ, and Hodges–Lehmann estimator with 95 % bootstrap confidence intervals (10,000 resamples), computed in R 4.3.1 (Sec. 3.6).
- **Systematic review:** 34 studies; 53 % include an empirical evaluation and none uses randomized controlled trials (Sec. 4.1).
- **Case selection:** based on willingness to participate, domain and stack diversity, established CI/CD infrastructure, and available baseline metrics (Sec. 3.3); the authors acknowledge a risk of selection bias.

3. Leakage Check

The study has no train/test split, but it is exposed to equivalent threats (temporal and metric contamination): the paper names them in Sec. 5.5 but does not test them. The most serious ones are listed below.

- **Censoring in defect density:** it is defined as production defects within 30 days (Table 2). With a 12-week treatment window, changes merged in the last four weeks were not observed for 30 days, which would bias density downward unless follow-up was extended (not mentioned).
- **Metrics affected by the intervention:** agents generate tests, so coverage and test executions rise by construction; about 48 % of pull requests are agent-authored, which also alters the KLOC denominator and PR counts (Sec. 5.2).
- **Unequal windows:** if the Sec. 4.2 totals are raw counts, the treatment period contains 50 % more weeks. Per week, commits rise 19 % (2,156 → 3,847), PRs 18 % (1,087 → 1,923), deployments 39 % (234 → 487), and test executions 14 % (52,109 → 89,234), while incidents fall 64 % (124 → 67), compared with raw increases of 78 %, 77 %, 108 %, and 71 % and a raw decrease of 46 %. The text does not clarify which case applies.
- **History and seasonality:** baseline and treatment fall in different periods; the statement that release cycles were "taken into account" is not substantiated, and there are no covariates or interrupted time series.
- **Novelty and learning effects:** volunteer teams, with gains concentrated in weeks 2–6, and no washout period or blinded evaluation.
- **Independence:** the authors designed, executed, and analyzed the study (Author Contributions); there is no preregistration or independent analysis.
- **Framework validated on the same cases:** the framework is proposed and "validated" on the same four projects; it is unclear whether it was adjusted during the study, and no holdout set exists.

4. Statistical Rigor

The reported p-values (0.005–0.02) cannot derive from a Wilcoxon signed-rank test on four teams: with $n = 4$ pairs, the minimum exact two-sided p-value is $2/16 = 0.125$.

- **Unit of analysis not stated.** If the units are teams, $p \geq 0.125$. If they are weeks, the 8 baseline and 12 treatment weeks do not form pairs (a Mann–Whitney test would apply), and weeks within a team are autocorrelated (pseudoreplication).
- **Bootstrap with four clusters:** resampling four teams yields unstable intervals, whereas resampling weeks ignores the team effect.
- **Multiple comparisons:** six outcomes are tested at $\alpha = 0.05$ without correction and without a power analysis (the authors only acknowledge limited power, Sec. 5.5).
- **Effect-size labels:** Sec. 3.6 sets "large" at $|\delta| \geq 0.47$, yet Table 4 labels $\delta = 0.58$ (defect density) and $\delta = 0.54$ (change failure rate) as "medium."

- **Aggregate medians only:** the paper claims improvement "consistent across all projects" without reporting per-project results.
- **Figure 6:** the three series appear as straight lines between the baseline and treatment medians, whereas the text states that gains concentrate in weeks 2–6 and then stabilize; it appears to be an illustrative trend rather than observed weekly data (visual inference).

The abstract and Table 4 also contradict each other:

Metric	Abstract	Table 4 / Sec. 4.3
Cycle time / lead time	−40 to −55 %; median −2.4 d, CI [−3.1, −1.7], p = 0.01	Only "lead time": 7.1 → 4.0 d, −3.1 d, CI [−4.0, −2.2], p = 0.008 (the text and Fig. 5 call 7.1 → 4.0 "cycle time")
MTTR	−29 min, CI [−38, −20], p = 0.02	−32 min, CI [−43, −21], p = 0.010
Test coverage	+11 to +45 % (the Discussion says "percentage points")	+18 pp (51 → 69)
Change failures	−7 to −11 %	−8 pp (24 → 16)

5. Missing Information

Information needed for an independent party to replicate the study is absent: the agent platform, the weekly data, and the review protocol.

- **Agent platform:** product, LLM and version, prompts and configuration, autonomy level, human-approval rules, and Open Policy Agent policies.
- **Study protocol:** preregistration, dates per project, how windows were aligned with release cycles, unit of analysis, missing weeks, and adoption curve per team.
- **Data:** per-project, per-week values; extraction and analysis scripts; and a data availability statement. The distinction between *lead time* and *cycle time*, which are used interchangeably, is also unclear.
- **Defined but unreported outcomes:** flaky-test rate, secrets incidents, cloud cost, pipeline duration, and perception surveys (Table 2 and Sec. 3.5); "escaped defects" and "time to first review" appear in the text without definition or figures.
- **Ethics:** no approval or consent is reported for developer data and surveys, and the organization is neither identified nor stated to be anonymized.
- **Systematic review:** databases, search strings, inclusion criteria, screening, and quality appraisal are not reported, and PRISMA 2020 (Page et al., 2021) is not cited. The number of records identified before the 34 included studies is not given, and the period changes: 2024–2025 (abstract) versus 2024–2026 (Sec. 2).
- **Computing environment:** only R 4.3.1 is reported; no packages or bootstrap seed.
- **Effect of governance:** there is no arm with AI but without governance, so the improvements cannot be attributed to governance rather than to the agent.

6. Reproducibility Scorecard

Result: 1 Yes, 5 Partial, 3 No. The machine-learning items were adapted to the quasi-experimental design (see the equivalences in Sections 3 and 4).

Item	Y / P / N	Evidence in the paper
Seeds reported, many runs	**Partial**	20 weeks × 4 teams and results with confidence intervals (Table 4), but medians without dispersion, no repeated runs of the (non-deterministic) agent, and no bootstrap seed.
Splits described	**Partial**	Baseline of 8 weeks and treatment of 12 weeks, team sizes, and stack (Table 1, Sec. 3.4); dates, unit of analysis, and alignment with release cycles are missing; windows are unequal.
Leakage checked	**No**	Threats are named (Sec. 5.5) but not tested; 30-day censoring in defect counts and metrics influenced by the intervention.
Tuning separated	**Partial**	The framework derives from the literature before the pilot (Sec. 3.1); it is not stated whether it was adjusted during the four projects, and there is no holdout set.
Statistical test named	**Partial**	Wilcoxon, Hodges–Lehmann, Cliff's δ, R 4.3.1 (Sec. 3.6); with n = 4 pairs the minimum p-value is 0.125, and no correction for multiple comparisons is applied.
Effect size and CI	**Yes**	Cliff's δ and 95 % CI per metric (Table 4), although the effect labels and the abstract's figures are inconsistent.
External validation	**No**	One organization, four projects, no replication; the authors acknowledge this (Sec. 5.5).
Compute documented	**Partial**	Pipeline tools and R 4.3.1; the agent platform, model, and versions are not reported.
Code and data available	**No**	No repository or statement; only aggregate medians (Table 4).

7. Overall Score

3.5 / 9 (≈ 3.9 / 10): low reproducibility. The paper is transparent about its limits, but today a third party could neither replicate it nor verify its figures.

Criterion: Yes = 1, Partial = 0.5, No = 0, unweighted (1 + 5 × 0.5 + 0 = 3.5).

- **Strengths:** precisely defined variables (Table 2), an 8-week baseline, nonparametric tests with effect size and confidence intervals, a description of the tool stack, and a candid validity section (Sec. 5.5).
- **Weaknesses:** no code or data, an unidentified agent platform, p-values incompatible with n = 4, figures that do not match between the abstract and the table, and a single organization.
- **Interpretation:** its conclusions are indicative, as the authors themselves acknowledge; they do not constitute evidence that agentic AI improves performance, still less that governance does.

8. Recommendations

1. **Publish data, scripts, and protocol.** Share anonymized per-project, per-week values, the R script with its seed, and the dates of each window; reconcile the abstract, Table 4, and Figures 5–6 (lead time versus cycle time, MTTR, percentage points); and report all twelve variables or justify the omissions. This improves the code-and-data, repeated-runs, and effect-size items.
2. **Redo the analysis with the correct unit.** Use an interrupted time series or a mixed-effects model with team as a random effect and weeks as observations, normalize counts per week, extend follow-up beyond 30 days for defects, correct for multiple comparisons, and avoid relying on a signed-rank test with n = 4. This improves the leakage and statistical-test items.
3. **Document the intervention and add a comparison.** Describe the platform, model, versions, configuration, and policies; include teams without AI or a staggered adoption to separate the effect of the agent, the

governance, and novelty; replicate in another organization; and report the review following PRISMA 2020. This improves the compute and external-validation items.

References

Alenezi, M., Akour, M., & Al-Omari, O. (2026). Agentic artificial intelligence in software engineering: Integration points, architectural frameworks, empirical evaluation, and governance. *International Journal of Technology, 17*(4), 1245–1262. https://doi.org/10.14716/ijtech.v17i4.8380

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